The change in magnetic flux in time dt is

$d\Phi_B = B \times$ change in area

$= B \times$ Area ABCD

$= Blvdt$ since Area ABCD $= l(vdt)$

(or) $\frac{d\Phi_B}{dt} = Blv$

As a result of change in flux, an emf is generated in the loop. The magnitude of the induced emf is

$$\varepsilon = \frac{d\Phi_B}{dt}$$

$$\varepsilon = Blv \qquad (4.27)$$

This emf is called **motional emf**. The direction of induced current is found to be clockwise from Fleming's right hand rule.

EXAMPLE 4.14

A circular metal of area 0.03 m^2 rotates in a uniform magnetic field of 0.4 T. The axis of rotation passes through the centre and perpendicular to its plane and is also parallel to the field. If the disc completes 20 revolutions in one second and the resistance of the disc is 4 Ω, calculate the induced emf between the axis and the rim and induced current flowing in the disc.

Solution

$A = 0.03$ m^2; $B = 0.4$ T; $f = 20$ rps;
$R = 4\ \Omega$

Area covered in 1 sec = Area of the disc $\times$ frequency

$= 0.03 \times 20$
$= 0.6$ m^2

Induced emf, ε= Rate of change of flux

$$= \frac{d\Phi_B}{dt} = \frac{d(BA)}{dt}$$

$$\varepsilon = \frac{0.4 \times 0.6}{1}$$

$$\varepsilon = 0.24 \text{ V}$$

Induced current, $i = \frac{\varepsilon}{R} = \frac{0.24}{4} = 0.06\,A$

Note

Emf can be induced by changing relative orientation between the coil and the magnetic field. This can be achieved either by rotating a coil in a magnetic field or by rotating a magnetic field within a stationary coil. Here rotating coil type is considered.

4.4.4 Induction of emf by changing relative orientation of the coil with the magnetic field

Consider a rectangular coil of N turns kept in a uniform magnetic field $\vec{B}$ as shown in Figure 4.25(a). The coil rotates in anti-clockwise direction with an angular velocity ω about an axis, perpendicular to the field.

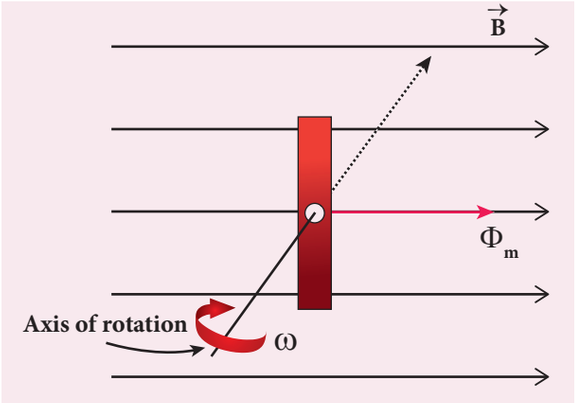


Figure 4.25(a) Top view of the coil with its plane perpendicular to the magnetic field

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